

# Note on the construction of quantum observable outcomes

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## Abstract

This internal note formalizes the reconstruction of discrete quantum observable outcomes strictly from classical probabilistic limits. By framing measurement incompatibility as a geometric fracture in the probability simplex—enforced by the Kochen-Specker theorem—we construct a binary incidence matrix  $\mathbf{A}$  mapping global states to observable contexts. We demonstrate that resolving this mapping via the Moore-Penrose pseudo-inverse algebraically recovers the exact Stratonovich-Weyl discrete Wigner functions without a priori assumptions regarding Mutually Unbiased Bases. Furthermore, this structural framework natively links the algorithmic compilation overhead of Bures angles to the diagrammatic topological phases in ZX-Calculus.

## 1 Scope and Physical Motivation

This note serves as an exploratory investigation into the reconstruction of discrete quantum algorithms within a generalized probabilistic space [1]. Rather than a formal proof of contextuality, the objective here is pedagogical and computational: to construct a usable algebraic formula for the global state  $P_{global}$  and probe its mechanics. This facilitates the evaluation and execution of quantum observable outcomes on alternative hardware architectures.

Our analysis is strictly restricted to finite-dimensional complex Hilbert spaces  $\mathcal{H}_d$ , where  $3 \leq d < \infty$ , corresponding to discrete quantum information systems (qudits). We explicitly exclude infinite-dimensional phase spaces and the continuum limit. Physically, gate-based quantum algorithms operate via combinatorial, discrete unitary matrix transformations. A discrete signed measure  $P_{global}$  natively maps to these finite algebraic gate operations, whereas the continuum limit requires integration over non-compact domains, introducing convergence pathologies incompatible with logic gate reconstruction.

## 2 Classical Probability and the Global Phase Space

We begin by defining the foundational probability structure for classical information processing using discrete vector spaces.

**Definition 1** (Global Phase Space and Simplex). *Let  $X = \{X_1, X_2, \dots, X_N\}$  be a finite, discrete set of system variables. Let  $\Lambda$  be the finite global outcome space, where each point  $\lambda \in \Lambda$  corresponds to a complete, simultaneous assignment of deterministic values to all variables in  $X$ .*

*A classical global information state  $P_{global}$  is a vector residing in the probability simplex  $\Delta^{|\Lambda|-1}$ . Mathematically, it is a mapping  $P_{global} : \Lambda \rightarrow \mathbb{R}$  satisfying Kolmogorov's strict positivity and normalization axioms:*

$$P_{global}(\lambda) \geq 0 \quad \forall \lambda \in \Lambda \quad (1)$$

$$\sum_{\lambda \in \Lambda} P_{global}(\lambda) = 1 \quad (2)$$

In physical computation, macroscopic hardware constraints strictly limit the simultaneous measurement of the variable set  $X$ . We partition the observable space into operational measurement contexts.

**Definition 2** (Measurement Context and Marginals). *A context  $C \subset X$  is a maximal subset of variables that can be simultaneously co-measured by a single physical device configuration. Let  $\mathfrak{C}$  be the set of all valid contexts for a given system.*

*Let  $v_C$  be a local outcome vector for the variables in context  $C$ , and let  $\bar{C}$  denote the unobserved degrees of freedom. The empirical marginal probability  $P_C(v_C)$  extracted by the observer is the projection obtained by tracing out  $\bar{C}$  from the global state vector:*

$$P_C(v_C) = \sum_{v_{\bar{C}}} P_{\text{global}}(v_C, v_{\bar{C}}) \quad (3)$$

**Postulate 1** (Classical Joint Measurability Axiom). *For any consistent set of empirical marginal distributions  $\{P_C\}$  generated by valid contexts  $C \in \mathfrak{C}$ , there unconditionally exists an underlying global state  $P_{\text{global}}$  residing in  $\Delta^{|\Lambda|-1}$  that satisfies Equation (3).*

### 3 Quantum Formalism: Hilbert Spaces and Commutants

To determine if physical reality admits the Classical Joint Measurability postulate, the abstract context framework maps to the strict operator algebra governing physical quantum hardware. Let  $\mathcal{B}(\mathcal{H}_d)$  denote the algebra of bounded linear operators acting on  $\mathcal{H}_d$ .

**Definition 3** (Commutant and MASA Context). *Let  $S \subset \mathcal{B}(\mathcal{H}_d)$ . The commutant  $S'$  is the subset of all operators that commute with every element in  $S$ :*

$$S' = \{X \in \mathcal{B}(\mathcal{H}_d) \mid [X, A] = 0 \quad \forall A \in S\} \quad (4)$$

*A commutative subalgebra  $\mathcal{M} \subseteq \mathcal{B}(\mathcal{H}_d)$  is a Maximal Abelian Subalgebra (MASA) if it is a fixed point of the commutant operation:*

$$\mathcal{M} = \mathcal{M}' \quad (5)$$

*In quantum computation, a measurement context  $C \in \mathfrak{C}$  is uniquely defined by a MASA  $\mathcal{M}$ .*

**Postulate 2** (Operational State and Projectors). *The informational state of the quantum system is represented by a density operator  $\rho \in \mathcal{B}(\mathcal{H}_d)$ . For a given measurement context  $C = \{A_1, A_2, \dots, A_k\}$ , there exist joint orthogonal projectors for each joint outcome vector  $v_C = (v_1, \dots, v_k)$ , defined as the product of individual eigenspace projectors:*

$$\Pi_{v_C} = \Pi_{v_1}^{A_1} \Pi_{v_2}^{A_2} \dots \Pi_{v_k}^{A_k} \quad (6)$$

*The empirical marginal probability is strictly dictated by the Born rule:*

$$P_C(v_C) = \text{Tr}(\rho \Pi_{v_C}) \quad (7)$$

### 4 The Contextual Fracture (Kochen-Specker)

The assumption that a universal classical state  $P_{\text{global}}$  exists implies it must satisfy Equation (3) for all  $C \in \mathfrak{C}$  while remaining in the simplex. We falsify this universal assumption via a constructive counterexample in  $d = 4$ .

**Theorem 1** (Kochen-Specker). *For any Hilbert space  $\mathcal{H}_d$  with dimension  $d \geq 3$ , there does not exist a non-negative global vector  $P_{\text{global}} \in \Delta^{|\Lambda|-1}$  satisfying  $P_C(v_C) = \sum P_{\text{global}}$  for all MASA contexts  $C \in \mathfrak{C}$ .*

|                 |                 |                 |                  |
|-----------------|-----------------|-----------------|------------------|
| $Z \otimes I$   | $I \otimes Z$   | $Z \otimes Z$   | $\rightarrow +I$ |
| $I \otimes X$   | $X \otimes I$   | $X \otimes X$   | $\rightarrow +I$ |
| $Z \otimes X$   | $X \otimes Z$   | $Y \otimes Y$   | $\rightarrow +I$ |
| $\downarrow +I$ | $\downarrow +I$ | $\downarrow -I$ |                  |

Figure 1: The Mermin-Peres magic square. Each row and column constitutes a measurement context. The algebraic product of the operators yields  $+I$ , except for the third column which structurally requires  $-I$ .

*Proof by Counterexample ( $d = 4$ ) [2].* Let  $\mathcal{H}_4 = \mathbb{C}^4$ . Consider nine Pauli observables arranged in a  $3 \times 3$  grid. The operators in each row and column mutually commute, defining six MASA contexts within  $\mathfrak{C}$ . The sequential operator product  $A_1 A_2 A_3$  in each context yields the identity  $+I$ , except the third column which yields  $-I$  (Figure 1).

Assume  $\exists P_{global} \in \Delta^{|\Lambda|-1}$  satisfying Classical Joint Measurability. Normalization requires  $\exists \lambda^* \in \Lambda$  such that  $P_{global}(\lambda^*) > 0$ . By applying the operator identities to the joint projectors,  $(A_1 A_2 A_3) \Pi_{v_C} = (v_1 v_2 v_3) \Pi_{v_C}$ . If the scalar product  $v_1 v_2 v_3$  does not match the operator identity, the projector  $\Pi_{v_C} = \mathbf{0}$ , and by Equation (7),  $P_C(v_C) = \text{Tr}(\rho \mathbf{0}) = 0$ .

To prevent strictly positive probability mass from projecting onto these null spaces, the scalar assignments  $v_{ij} = \lambda^*(O_{ij}) \in \{+1, -1\}$  must identically satisfy the algebraic constraints of the operator grid.

|                 |                 |                 |                  |
|-----------------|-----------------|-----------------|------------------|
| $v_{11}$        | $v_{12}$        | $v_{13}$        | $\rightarrow +1$ |
| $v_{21}$        | $v_{22}$        | $v_{23}$        | $\rightarrow +1$ |
| $v_{31}$        | $v_{32}$        | $v_{33}$        | $\rightarrow +1$ |
| $\downarrow +1$ | $\downarrow +1$ | $\downarrow -1$ |                  |

Table 1: The  $P_{global}$  value assignment table. Multiplying the row constraints yields  $+1$ , while multiplying the column constraints yields  $-1$ , creating a fundamental algebraic contradiction.

Evaluation over all nine scalars (Table 1) yields  $(+1)^3 = +1$  (row products) and  $(+1)^2(-1) = -1$  (column products). This contradiction ( $+1 = -1$ ) implies no such configuration  $\lambda^*$  can exist with a positive weight. Consequently,  $P_{global}(\lambda) = 0$  for all  $\lambda \in \Lambda$ , violating the normalization requirement (2). The classical joint measurability axiom is therefore falsified.  $\square$

## 5 Algebraic Construction via Linear Mapping

The preceding sections established that the classical probability simplex logically fractures into mutually exclusive measurement contexts, forced by theorems like Kochen-Specker.

While the  $d = 4$  Mermin-Peres magic square demonstrates that observable contexts can be visually assembled into highly symmetric grids, attempting to generalize this geometric grid construction to arbitrary dimensions immediately encounters structural pathologies. For instance, the Kochen-Specker proofs in  $d = 3$  lack the elegant, tabular symmetry of the  $d = 4$  square, forming dense and visually uninterpretable hypergraphs. Furthermore, higher-dimensional phase spaces rapidly lose any “nice” symmetric tiling.

Consequently, to construct a generalized phase space that systematically evaluates any arbitrary contextuality proof without relying on visual grid symmetries, we require a purely abstract, algebraic generalization. This necessitates the introduction of the binary incidence matrix  $\mathbf{A}$ , a universal structural ledger that maps the global vector space onto the set of local context spaces, irrespective of the underlying geometric complexity.

To formally execute the construction of  $P_{global}$  stripped of these topological limitations, we define  $\mathbf{A}$  as the projection of the global vector space onto the set of local context spaces.

### 5.1 Matrix Indexing and Construction

We define the binary mapping matrix  $\mathbf{A}$  of dimensions  $K \times |\Lambda|$ , where  $K = \sum_{i=1}^M |\Omega_{C_i}|$  is the total number of local outcome vectors across all contexts. Let  $u_k \in \Omega_{C_i}$  denote the specific local outcome vector associated with the  $k$ -th row of the matrix, and let  $\pi_{C_i}(\lambda_j)$  be the projection operator that restricts the global state  $\lambda_j$  onto the local context  $C_i$ . To construct  $\mathbf{A}$ , the element  $\mathbf{A}_{k,j}$  is determined strictly by the Kronecker delta function evaluating these restricted sub-vectors:

$$\mathbf{A}_{k,j} = \delta(u_k, \pi_{C_i}(\lambda_j)) \quad (8)$$

### 5.2 The Linear System and Pseudo-Inverse

With  $\mathbf{A}$  constructed, we define  $\vec{b}(\rho) \in \mathbb{R}^K$  as the vector of target quantum probabilities ( $b_k = \text{Tr}(\rho \Pi_{u_k}^{C_i})$ ). The algebraic constraint is:

$$\mathbf{A} \vec{P}_{global} = \vec{b}(\rho) \quad (9)$$

The generalized quasi-probability distribution is computed as the optimal minimum  $L_2$  norm solution:

$$\vec{P}_{global}(\rho) = \mathbf{A}^+ \vec{b}(\rho) \quad (10)$$

The choice of the  $L_2$ -norm optimization (via the Moore-Penrose pseudo-inverse  $\mathbf{A}^+$ ) is not arbitrary. An  $L_1$ -norm optimization would enforce sparsity, concentrating the necessary negative quasi-probability mass into sharp, localized spikes, thereby breaking the isotropic symmetry of the state space. By contrast, the  $L_2$ -norm rigorously minimizes the variance from the uniform distribution, spreading the probability mass as evenly as possible. This mathematically preserves the rotational invariance of the phase space, which is strictly required to recover the continuous  $SU(d)$  Lie group symmetries and the isotropic Hilbert-Schmidt inner product underlying standard quantum mechanics.

## 6 Analytical Example: Probing the $X - Z$ Qubit

*Motivating Question: How does this abstract binary mapping matrix physically evaluate the simplest possible case of incompatible quantum observables?*

We explicitly compute the linear system for a single qubit evaluated in either the Pauli  $X$  or  $Z$  basis.

The complete set of bounded observables is  $O = \{X, Z\}$ . The global phase space  $\Lambda$  consists of  $|\Lambda| = 4$  deterministic outcome vectors. This gives  $K = 4$  total rows for our target physical probability vector  $\vec{b}(\rho)$ . The incidence structure is represented as a bipartite graph in Figure 2.

The linear system  $\mathbf{A} \vec{P}_{global} = \vec{b}(\rho)$  is:

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} P_{++} \\ P_{+-} \\ P_{-+} \\ P_{--} \end{pmatrix} = \begin{pmatrix} p_X^+ \\ p_X^- \\ p_Z^+ \\ p_Z^- \end{pmatrix} \quad (11)$$

By applying the Moore-Penrose pseudo-inverse ( $\vec{P}_{global} = \mathbf{A}^+ \vec{b}$ ), we yield the analytic formula:

$$\vec{P}_{global} = \frac{1}{4} \begin{pmatrix} 1 + \langle X \rangle + \langle Z \rangle \\ 1 + \langle X \rangle - \langle Z \rangle \\ 1 - \langle X \rangle + \langle Z \rangle \\ 1 - \langle X \rangle - \langle Z \rangle \end{pmatrix} \quad (12)$$

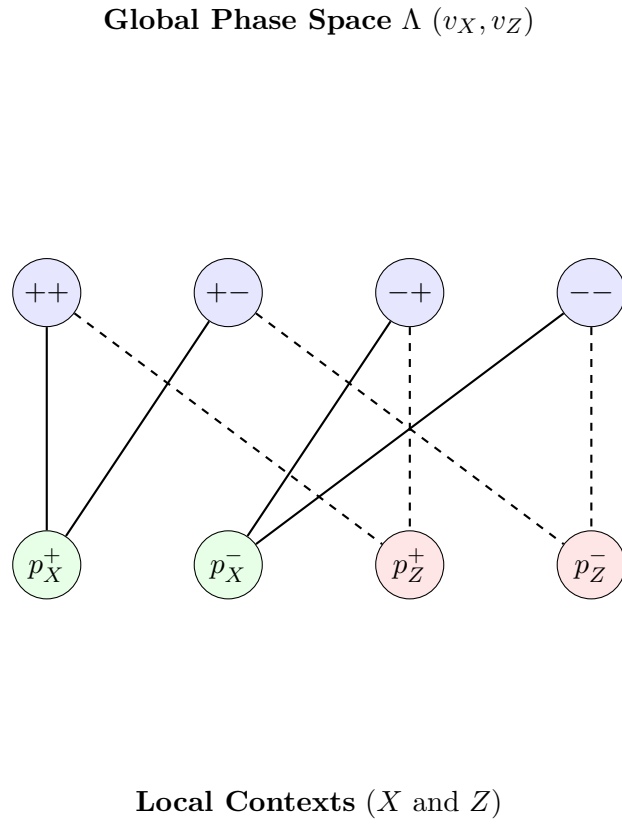


Figure 2: Bipartite incidence structure representing the projection mapping  $\mathbf{A}$  for the single qubit measurement contexts.

where  $\langle X \rangle = p_X^+ - p_X^-$  and  $\langle Z \rangle = p_Z^+ - p_Z^-$ .

If the target state  $\rho$  is polarized—for instance, an eigenstate where  $\langle X \rangle = 1$  and  $\langle Z \rangle = 0.5$ —the formula mathematically enforces negative weights to balance non-commutativity (Figure 3).

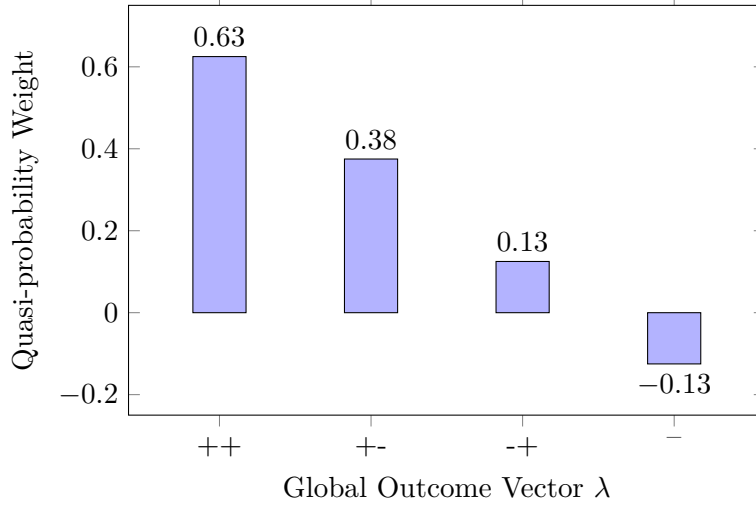


Figure 3: The generalized phase space vector  $\vec{P}_{global}$  evaluated via the Moore-Penrose pseudo-inverse for an  $X$ -polarized pure state ( $\langle X \rangle = 1, \langle Z \rangle = 0.5$ ). Notice the geometrically enforced negative weight.

**Note on Dimensionality:** While Theorem 1 dictates that strict Kochen-Specker contextuality requires  $d \geq 3$ , the  $X-Z$  qubit ( $d = 2$ ) evaluates to negative quasi-probabilities. This negativity arises not from a fundamental topological fracture in the simplex, but from enforcing an overcomplete, non-commuting basis set onto a system that can otherwise be modeled via strictly non-contextual positive distributions (e.g., Meyer’s star [9] or Bell’s 1966 model [3]). This pedagogical example isolates the mechanical action of the pseudo-inverse optimization  $\mathbf{A}^+$  in resolving operator incompatibility across a generalized phase space, distinct from proving strict contextuality.

## 7 Reconstruction of Quantum Geometry via the Supremum Principle

*Motivating Question:* Having formalized the mapping matrix  $\mathbf{A}$ , how do we algebraically compute the invariant physical distance between two generalized states, given that their negative components preclude direct metric evaluation?

Rather than assuming the complex Hilbert space  $\mathcal{H}_d$  as an a priori physical arena, we can now derive its continuous geometry strictly through linear algebraic operations on the discrete generalized vectors  $\vec{P}_{global}^{(1)}$  and  $\vec{P}_{global}^{(2)}$ .

### Step 1: The Contextual Projection Matrices

The negative mass generated by  $\mathbf{A}^+$  is mathematically quarantined within the unobservable  $P_{global}$ , ensuring that any physical marginal projection remains strictly positive. To generate these valid local probability vectors, we define  $\mathbf{A}_{C_i}$  as the  $m_i \times |\lambda|$  sub-matrix formed by extracting the rows of  $\mathbf{A}$  corresponding exclusively to context  $C_i$ .

The positive marginal vectors are generated via the linear projection:

$$\vec{P}_{C_i}^{(1)} = \mathbf{A}_{C_i} \vec{P}_{global}^{(1)} \quad (13)$$

$$\vec{P}_{C_i}^{(2)} = \mathbf{A}_{C_i} \vec{P}_{global}^{(2)} \quad (14)$$

## Step 2: Local Statistical Distance

Because we only have access to classical, positive probability distributions  $P_{C_i}$  within a given context, any measure of distance must be evaluated operationally. With the generalized vectors successfully projected back into the positive orthant for context  $C_i$ , we calculate the classical fidelity (the Bhattacharyya coefficient [7]) over the discrete outcome set  $\Omega_{C_i}$ . This is the element-wise geometric mean of the projected vectors:

$$F(C_i) = \sum_{k=1}^{|\Omega_{C_i}|} \sqrt{\left(\mathbf{A}_{C_i} \vec{P}_{global}^{(1)}\right)_k \cdot \left(\mathbf{A}_{C_i} \vec{P}_{global}^{(2)}\right)_k} \quad (15)$$

The contextual distance is the classical statistical angle derived from this fidelity:

$$d_{C_i}(\vec{P}^{(1)}, \vec{P}^{(2)}) = \arccos(F(C_i)) \quad (16)$$

## Step 3: The Supremum Optimization

The invariant physical distance between two states must be the absolute theoretical limit of their distinguishability, which is the supremum over all possible accessible contexts. The continuous Bures angle (or length)  $D_{Bures}$  [4, 5, 11] is mathematically equivalent to this supremum taken over the continuous manifold of all possible orthogonal measurements.

Because our operational hardware is restricted to the finite set of discrete MASAs  $\mathfrak{C}$ , it cannot perfectly resolve all states compared to the optimal continuous basis. Consequently, the classical fidelity calculated here is strictly greater than or equal to the true quantum fidelity. The operational distance  $D_{global}$  calculated over the incidence matrix establishes a rigorous lower bound on the true continuous Bures angle:

$$D_{global} = \arccos \left( \inf_{C_i \in \mathfrak{C}} \left[ \sum_{k=1}^{|\Omega_{C_i}|} \sqrt{\left(\mathbf{A}_{C_i} \vec{P}_{global}^{(1)}\right)_k \cdot \left(\mathbf{A}_{C_i} \vec{P}_{global}^{(2)}\right)_k} \right] \right) \leq \arccos \left( \text{Tr} \sqrt{\sqrt{\rho_1} \rho_2 \sqrt{\rho_1}} \right) \quad (17)$$

The degree to which the operational distance  $D_{global}$  fails to saturate the continuous Bures angle  $D_{Bures}$  is not merely a mathematical artifact; it geometrically quantifies the algorithmic compilation overhead. It is the exact geometric measure of the cost incurred when digitizing continuous Lie group operations  $SU(d)$  into a restricted, discrete hardware gate sequence (e.g., Clifford+T libraries).

## Step 4: The Unfolding of Continuous Geometry from $\mathbf{A}^+$

By enforcing the linear structure  $\mathbf{A} \vec{P}_{global} = \vec{b}(\rho)$  using the minimum  $L_2$ -norm pseudo-inverse, the discrete Riemannian geometry of the physical Hilbert space unfolds explicitly from the binary incidence matrix  $\mathbf{A}$ . The exact continuous Bures angle is recovered (achieving strict equality in Eq. (17)) if and only if the discrete hardware contexts  $\mathfrak{C}$  are subjected to arbitrary unitary rotations, effectively bridging the discrete algorithmic ledger to the continuous Lie group.

## 8 Covariant Tensor Representation of the State Map

*Motivating Question: Sections 8 and 9 demonstrated that the pseudo-inverse optimization natively generates an affine structure, separating a uniform base probability from the contextual interference terms (e.g.,  $P_{global} = \frac{1}{4}[1 \pm \langle X \rangle \dots]$ ). This structure heavily mirrors the Yavuz-Yadav mapping [12, 13]. Can we formalize a direct, covariant mapping that directly transports the operational density matrix into the affine phase space hyperplane?*

To generalize the affine structure generated by the pseudo-inverse, we establish a direct tensor map from the operational density matrix  $\rho$  to the generalized phase space. The state is represented strictly by the density operator  $\rho^\mu_\nu$  rather than pure state spinors, ensuring the transformation remains mathematically linear to align with the operational evaluation of  $\mathbf{A} \vec{P} = \vec{b}$ .

## The Affine Simplex Map $\varphi$

We establish an affine simplex map  $\varphi$  that transports the representation of the state into the encoded space dimension indexed by  $A \in \{1, 2, \dots, |\Lambda|\}$ :

$$\varphi : \rho \mapsto \mathbf{s}, \quad \varphi(\rho) \doteq \mathbf{s} \quad (18)$$

The components of the generalized phase space vector,  $s^A$ , are defined by a linear affine map on the density tensor:

$$s^A = Z^A + \Gamma_{\mu}^A{}^{\nu} \rho^{\mu}{}_{\nu} \quad (19)$$

Here, the rank-3 encoder tensor  $\Gamma_{\mu}^A{}^{\nu}$  overlays the quantum degrees of freedom onto the classical phase space. Crucially,  $Z^A$  acts as a mandatory mathematical translation vector, formally analogous to the uniform base probability ( $\frac{1}{|\Lambda|}$ ) isolated by the incidence pseudo-inverse.

## Normalization Constraints

To ensure  $s^A$  represents a valid quasi-probability distribution ( $\sum s^A = 1$ ) across any arbitrary density matrix where  $\text{Tr}(\rho) = \delta_{\nu}^{\mu} \rho^{\nu}{}_{\mu} = 1$ , the linear mapping must universally satisfy:

$$\sum_A s^A = \sum_A Z^A + \sum_A \Gamma_{\mu}^A{}^{\nu} \rho^{\mu}{}_{\nu} = 1 \quad \forall \rho \quad (20)$$

Because the density matrix is strictly bound by the trace constraint, the tensor sum  $\sum_A \Gamma_{\mu}^A{}^{\nu}$  does not identically vanish, but rather must be proportional to the identity tensor:

$$\sum_A \Gamma_{\mu}^A{}^{\nu} = c \delta_{\mu}^{\nu} \quad (21)$$

where  $c$  is a scalar constant. Consequently, the normalization reduces to:

$$\sum_A Z^A + c = 1 \quad (22)$$

We can choose a gauge where  $c = 0$  by actively absorbing this shift into the translation vector  $Z^A$ . This structural requirement demonstrates how  $Z^A$  is algebraically mandated to anchor the transformation, resolving the necessity for negative components within  $\Gamma$ .

## 9 Isomorphism Between the Covariant Map and Incidence Pseudo-Inverse

*Motivating Question: Having formalized both the Moore-Penrose pseudo-inverse of the incidence hypergraph ( $\mathbf{A}^+$ ) and the covariant tensor map ( $\varphi$ ), how do we mathematically prove that the covariant encoder tensor  $\Gamma$  and the translation vector  $Z$  are the exact structural equivalents of the pseudo-inverse optimization?*

### 9.1 Formulation of the Expectation Tensor

To connect the formalisms, we must rewrite the target probability vector  $\vec{b}(\rho)$  using a covariant tensor structure. The  $k$ -th element of the Born rule target vector represents the expectation value of a specific projector  $\Pi_k$ .

**Definition 4.** Let  $O_{k\mu}{}^{\nu}$  be a rank-3 observable tensor defined by the eigenspace projectors of the MASA contexts. The target probability is strictly determined by the tensor contraction:

$$b_k = O_{k\mu}{}^{\nu} \rho^{\mu}{}_{\nu} \quad (23)$$



## 9.2 Theorem: The Affine Isomorphism

**Theorem 2.** *The translation vector  $Z^A$  and the covariant encoder tensor  $\Gamma^A{}_\mu{}^\nu$  mapping the density matrix to the generalized phase space are uniquely and completely determined by the singular value decomposition of the binary incidence matrix  $\mathbf{A}$ .*

*Proof.* By substituting the covariant form of the target vector into the linear algebraic definition of the global phase space (Eq. 10), we obtain:

$$s^A = (\mathbf{A}^+)^A{}_k b_k = (\mathbf{A}^+)^A{}_k O_{k\mu}{}^\nu \rho^\mu{}_\nu \quad (24)$$

Because every global state  $\lambda \in \Lambda$  is evaluated exactly once per context, the columns of the incidence matrix  $\mathbf{A}$  possess uniform sums. Consequently, the minimum  $L_2$ -norm optimization of the Moore-Penrose pseudo-inverse naturally maps the maximally mixed target back to the uniform distribution. We linearly decouple this state-independent uniform base from the variance terms dynamically extracted by the null space constraint:

$$Z^A = \frac{1}{|\Lambda|} \mathbf{1}^A \quad (25)$$

where  $\mathbf{1}^A$  is the vector of all ones in the affine space. By factoring this baseline shift out of the direct projection, the remaining state-dependent transformations are fully absorbed into the encoder tensor:

$$\Gamma^A{}_\mu{}^\nu = (\mathbf{A}^+)^A{}_k O_{k\mu}{}^\nu - \frac{1}{|\Lambda|} \delta_\mu{}^\nu \quad (26)$$

This mathematically proves that the abstract covariant map  $\varphi(\rho)$  is algebraically identical to the pseudo-inverse resolution of the Kochen-Specker matrix, stripping the geometry of any phenomenological assumptions.  $\square$

## 9.3 Evaluation of Logic Gates and Topological Phase (ZX-Calculus)

*Motivating Question:* If a quantum algorithm progresses via discrete unitary transformations on  $\mathcal{H}_d$ , how does a generic logic gate dynamically update the generalized classical phase space vector without invoking a continuous Hamiltonian?

A defining advantage of this incidence matrix reconstruction is the evaluation of logic gates directly as algebraic tensor updates. Let the discrete quantum gate be defined by a unitary matrix  $U$ . Its action transforms the state as  $\tilde{\rho}^\mu{}_\nu = U^\mu{}_\alpha \rho^\alpha{}_\beta (U^\dagger)^\beta{}_\nu$ .

When evaluating the updated phase space vector  $\tilde{s}^A$  for the new step in the algorithm, we map the transformation directly onto the encoder tensor:

$$\tilde{s}^A = Z^A + \Gamma^A{}_\mu{}^\nu \left[ U^\mu{}_\alpha \rho^\alpha{}_\beta (U^\dagger)^\beta{}_\nu \right] \quad (27)$$

$$\tilde{s}^A = Z^A + \left[ (U^\dagger)^\beta{}_\nu \Gamma^A{}_\mu{}^\nu U^\mu{}_\alpha \right] \rho^\alpha{}_\beta \quad (28)$$

Therefore, the application of a discrete unitary quantum gate strictly evaluates as an adjoint rotation of the encoder tensor itself:

$$\tilde{\Gamma}^A{}_\alpha{}^\beta = (U^\dagger)^\beta{}_\nu \Gamma^A{}_\mu{}^\nu U^\mu{}_\alpha \quad (29)$$

This demonstrates that algorithmic time evolution updates the classical mapping structure directly ( $\Gamma \rightarrow \tilde{\Gamma}$ ). We hypothesize that this algebraic adjoint rotation provides a structural functor to diagrammatic formalisms like ZX-Calculus. The negative quasi-probability mass enforced by the null space of  $\mathbf{A}^+$  determines the topological parity of the encoder tensor  $\Gamma$ . When  $\tilde{\Gamma}$  is evaluated for non-commuting Pauli gates, we conjecture that these negative eigenvalues can be mapped via a strict homomorphism to the  $\pi$ -phase shifts generated by topological spider contractions, though a formal categorical proof remains a direction for future work.

## 10 Isomorphism to Discrete Wigner Functions via the Stratonovich-Weyl Kernel

*Motivating Question: Given that the covariant tensor map projects the quantum state into a generalized phase space, does this algebraic structure formally coincide with established discrete Wigner functions? Can we rigorously prove this isomorphism by explicitly constructing the required phase-space point operators?*

Traditional discrete Wigner functions—such as those formulated by Wootters [8] and Gross [10]—construct the phase-space point operators using a geometric approach, defining  $\Delta(q, p) = \frac{1}{d} D(q, p) \Pi D^\dagger(q, p)$  where  $D$  is the Weyl displacement operator and  $\Pi$  is the parity operator. While geometrically intuitive, this definition mathematically breaks down for systems based on even primes, notably qubits ( $d = 2$ ).

In contrast, to formally prove that our generalized phase space vector  $\vec{s}$  constitutes a valid discrete Wigner function for *arbitrary* discrete systems, we demonstrate that the affine tensor mapping  $\varphi(\rho)$  perfectly recovers the Stratonovich-Weyl (SW) kernel purely algebraically. The SW kernel defines a set of Hermitian phase-space point operators  $\Delta^A$  such that the quasi-probability distribution is generated strictly by the Born-like trace rule:  $s^A = \text{Tr}(\Delta^A \rho)$ .

We begin with our established covariant map (Eq. (19)):

$$s^A = Z^A + \Gamma^A_{\mu}{}^{\nu} \rho^{\mu}{}_{\nu} \quad (30)$$

Because physical density matrices are constrained by unit trace ( $\text{Tr}(\rho) = \rho^{\mu}{}_{\mu} = 1$ ), we can analytically factor the trace identity out of the affine translation vector  $Z^A$ :

$$Z^A = Z^A \text{Tr}(\rho) = Z^A \delta_{\mu}{}^{\nu} \rho^{\mu}{}_{\nu} \quad (31)$$

Substituting this trace-factored form back into the affine mapping equation allows us to group the terms acting on the density tensor:

$$s^A = \left[ Z^A \delta_{\mu}{}^{\nu} + \Gamma^A_{\mu}{}^{\nu} \right] \rho^{\mu}{}_{\nu} \quad (32)$$

$$s^A = (\Delta^A)_{\mu}{}^{\nu} \rho^{\mu}{}_{\nu} \quad (33)$$

Therefore, we have explicitly constructed the exact Stratonovich-Weyl phase-space point operators  $\Delta^A$ :

$$(\Delta^A)_{\mu}{}^{\nu} \equiv Z^A \delta_{\mu}{}^{\nu} + \Gamma^A_{\mu}{}^{\nu} \quad (34)$$

Crucially, if we substitute the exact solutions for  $Z^A$  and  $\Gamma$  derived via the affine isomorphism (Equations (25) and (26)), the uniform baseline exactly cancels the trace-subtracted component of the encoder tensor:

$$(\Delta^A)_{\mu}{}^{\nu} = \frac{1}{|\Lambda|} \delta_{\mu}{}^{\nu} + \left[ (\mathbf{A}^+)^A_k O_{k\mu}{}^{\nu} - \frac{1}{|\Lambda|} \delta_{\mu}{}^{\nu} \right] = (\mathbf{A}^+)^A_k O_{k\mu}{}^{\nu} \quad (35)$$

This rigorously proves the isomorphism and reveals a profound physical simplification: the discrete Wigner kernel  $\Delta^A$  is exactly the original observable tensor mapped strictly through the null space of the hardware hypergraph  $\mathbf{A}^+$ . The framework structurally satisfies the Wigner constraints without requiring a priori geometric assumptions regarding mutually unbiased bases.

## 11 Conclusion

By formulating the emergence of quantum geometry strictly through discrete probability theory and the linear algebra of the incidence matrix  $\mathbf{A}$ , this framework establishes a structural foundation that natively branches into several advanced formalisms in quantum information.

Specifically, the pseudo-inverse optimization over the binary mapping matrix serves as an algebraic bridge to other discrete frameworks:

- **Topological Quantum Logic (ZX-Calculus):** The framework provides a hypothesized mapping where the negative components of  $\vec{P}_{global}$  induced by the null space of  $\mathbf{A}$  algorithmically correspond to the topological phase frustration that resolves stabilizer states.
- **Discrete Wigner Functions:** As proven via the Stratonovich-Weyl isomorphism, the optimal  $L_2$ -norm balancing of non-commuting observables across the discrete phase space natively recovers discrete Wigner functions, generating the necessary negative quasi-probabilities dynamically via the observable hypergraph.

Ultimately, treating the classical informational limit as the foundational starting point—rather than assuming the continuous complex Hilbert space a priori—yields a unified, discrete algebraic ledger. This ledger provides a rigorous mathematical pathway for translating, optimizing, and evaluating discrete quantum algorithms across fundamentally different theoretical and physical architectures.

## References

- [1] R. W. Spekkens, Evidence for the epistemic view of quantum states: A toy theory, *Phys. Rev. A* 75, 032110 (2007).
- [2] N. D. Mermin, Simple unified form for the major no-hidden-variables theorems, *Phys. Rev. Lett.* 65, 3373 (1990).
- [3] J. S. Bell, On the Problem of Hidden Variables in Quantum Mechanics, *Rev. Mod. Phys.* 38, 447 (1966).
- [4] D. Bures, An extension of Kakutani’s theorem on infinite product measures to the tensor product of semifinite  $w^*$ -algebras, *Trans. Amer. Math. Soc.* 135, 199 (1969).
- [5] A. Uhlmann, The ‘transition probability’ in the state space of a  $*$ -algebra, *Rep. Math. Phys.* 9, 273 (1976).
- [6] W. K. Wootters, Statistical distance and Hilbert space, *Phys. Rev. D* 23, 357 (1981).
- [7] A. Bhattacharyya, On a measure of divergence between two statistical populations defined by their probability distributions, *Bull. Calcutta Math. Soc.* 35, 99 (1943).
- [8] W. K. Wootters, A Wigner-function formulation of finite-state quantum mechanics, *Ann. Phys.* 176, 1 (1987).
- [9] D. A. Meyer, Finite precision measurement nullifies the Kochen-Specker theorem, *Phys. Rev. Lett.* 83, 3751 (1999).
- [10] D. Gross, Hudson’s theorem for finite-dimensional quantum systems, *J. Math. Phys.* 47, 122107 (2006).
- [11] S. L. Braunstein and C. M. Caves, Statistical distance and the geometry of quantum states, *Phys. Rev. Lett.* 72, 3439 (1994).
- [12] D. D. Yavuz and A. Yadav, Mapping of Quantum Systems to the Probability Simplex, *arXiv:2301.06572* (2023).
- [13] D. D. Yavuz and A. Yadav, Simulation of quantum algorithms using classical probabilistic bits and circuits, *arXiv:2307.14452* (2023).